

Sales & Invoice Data Entry and Cleaning Project

1. Project Overview

This project focuses on cleaning and standardizing sales invoice data generated from manual data entry processes. The dataset consists of **1000 transaction records** containing common real-world data quality issues such as duplicate entries, inconsistent formatting, missing values, and typographical errors.

The objective of this project is to **transform raw, error-prone data into a clean, reliable, and analysis-ready dataset** by applying systematic data cleaning techniques in Google Sheets. The project demonstrates practical skills in data validation, data consistency enforcement, and documentation of before-and-after data conditions to support accurate reporting and business decision-making.

2. Dataset Summary

- Total rows (raw data): 1000
- Total columns: 12
- Data period: January – June 2024
- Format: Excel / Google Sheets
- Data type: Sales & Invoice Transactions

Key Features:

- Transaction details: `invoice_id`, `invoice_date`, `total_amount`
- Customer information: `customer_name`, `customer_type`, `city`
- Product information: `product_name`, `category`, `qty`, `unit_price`
- Payment status and sales channel

3. Data Cleaning Process Using Google Sheets

All data cleaning was performed in **Google Sheets**. The raw dataset was preserved in `raw_sales_data`, while all transformations were applied in a separate sheet, `clean_sales_data`, with each step documented for traceability.

Cleaning Steps & Formulas

- **Duplicate Invoice Removal**
Removed duplicate `invoice_id` values using
Data → Data cleanup → Remove duplicates.
- **Date Standardization**
To convert mixed and text-based date formats into a standardized date format while handling hidden spaces. Converted mixed date formats into a single standard format

(yyyy-mm-dd):

=IFERROR(DATEVALUE(CLEAN(TRIM(B2))),DATE(2000+RIGHT(B2,2), MID(B2,4,2), LEFT(B2,2)))

- **Customer Name Formatting**

Standardized inconsistent text casing:

=PROPER(D2)

- **Customer Type Correction**

Fixed typos and normalized categories:

=IF(LOWER(F2)="reseler","Reseller",PROPER(F2))

- **Product Name Cleaning**

Removed extra spaces from text entries:

=TRIM(H2)

- **Quantity Validation**

Because a zero quantity implies no transaction, while a blank value indicates missing data. To preserve the business meaning and avoid biasing sales metrics, I imputed missing quantities using the median quantity:

=IF(OR(K2="",K2=0),MEDIAN(FILTER(K:K,K:K>0)),K2)

- **Payment Status Standardization**

Standardized categorical values:

=IF(UPPER(A2)="PAID","Paid","Unpaid")

- **City Name Normalization**

Mapped inconsistent city names to standard values:

=IF(OR(A2="JKT",A2="Jakrta"),"Jakarta",A2)

- **Total Amount Recalculation**

Ensured accurate transaction values:

=qty_clean * unit_price_clean

- **Derived Sales Channel from Customer Type**

Sales channel values were derived using a business rule based on customer type to standardize channel classification:

=IF(G2="Reseller","Online",IF(G2="Individual","Direct","Unknown"))

- **Data Validation Rules**

Applied dropdowns and constraints for key fields using

Data → *Data validation*.

a.) Customer Type: Dropdown validation (Individual, Reseller) with input rejection

b.) Payment Status: Dropdown validation (Paid, Unpaid) with input rejection

c.) Quantity: Numeric validation (> 0) with input rejection

- **Conditional Formatting**

Applied conditional formatting to visually highlight payment status (**Paid in green, Unpaid in red**)

Format → Conditional formatting

4. Before vs After Cleaning Summary

After completing the cleaning process:

- Duplicate records were eliminated
- Date and text formats were standardized
- Missing and invalid values were resolved
- The dataset became fully analysis-ready

The cleaned dataset is now suitable for:

- Dashboard creation
- Sales analysis
- Financial reporting

5. Tools Used

- Google Sheets
 - Data Cleanup
 - Data Validation
 - Spreadsheet Functions

6. Key Skills Demonstrated

- Data Entry & Data Quality Control
- Data Cleaning & Validation
- Spreadsheet Functions & Logic

7. Conclusion

This project highlights the importance of data quality as the foundation of analytics.

Through a structured and well-documented cleaning process, raw manually entered data was transformed into a reliable dataset ready to support business decision-making.

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